Honors Numerical Analysis

MATH-UA 396

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Contents

0 Introduction **2**

1 IEEE Arithmetic **2**

1.1 Computer Representation of Numbers . . . 2

1.2 Rounding . . . 3

2 Solving Nonlinear Equations **3**

2.1 Bisection . . . 4

2.2 Secant Method . . . 4

2.3 Newton's Method . . . 4

3 Solving Nonlinear Systems **6**

3.1 Multivariable Vector-Valued Taylor Series . . . 8

3.2 Multivariate Newton's Method . . . 9

4 Optimization **9**

4.1 Single Variable Optimization . . . 9

4.2 Multivariate Optimization . . . 10

5 Numerical Linear Algebra **11**

5.1 Gaussian Elimination . . . 11

5.2 LU Factorization . . . 12

5.3 Cholesky Factorization . . . 13

5.4 Row Pivoting . . . 13

5.5 Conditioning and Backward Stability . . . 14

5.6 QR Factorization . . . 15

5.7 Gram-Schmidt (Triangular) Orthogonalization . . . 15

5.8 Householder Reflection (Orthogonal) Triangularization . . . 17

5.9 The Singular Value Decomposition . . . 19

6 Eigenvalue Problems **21**

6.1 Solving Systems of Linear Initial Value Problems . . . 21

6.2 The Gerschgorin Theorems 22
6.3 The Power Method 22
6.4 The Jacobi Method 24
6.5 The QR Method 27

1/23 Lecture

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Programming language. Julia, with `git` version control and Github Classroom.

Grading. Final 40%, Midterm 30%, Homework each 5%.

Course materials.

- Overton, *Numerical Computing with IEEE Floating Point Arithmetic*, SIAM, 2001.
- Driscoll and Braun, *Fundamentals of Numerical Computation*, SIAM, 2017.
- Trefethen and Bau, *Numerical Linear Algebra*, SIAM, 1997.
- Suli and Mayers, *An Introduction to Numerical Analysis*, Cambridge, 2003.

0 Introduction

- **Nonlinear Equations.** Nonlinear equations, nonlinear equations of several variables, and systems of nonlinear equations of several variables.
- **Linear Algebra.** Solving $A\vec{x} = \vec{b}$, eigenproblems $A\vec{v} = \lambda\vec{v}$, decompositions $A = QR = USV^\top$, and optimizations $y = \arg\min f(x)$.
- **Interpolation and Approximation.** Interpolants and approximation.
- **Numerical Integration.** Riemann integrals and quadratures.
- **Fast Algorithms.** Fast Fourier transform $O(n \log n)$ and Monte Carlos.

1/25 Lecture

1 IEEE Arithmetic

1.1 Computer Representation of Numbers

Fixed point and floating point representation. A fixed point representation specifies a fix number of bits for the integer and fraction part. For instance, with specified format "XXXX.XXXX", 7.5 becomes 0007.5000. A floating point representation, on the other hand, is represented $\pm M \times 2^E$, where $M \in [1, 2)$ is called the mantissa and E is called the exponent. For instance, $23 = (10111)_2 = (1.0111)_2 \times 2^4$, $0.125 = (0.001)_2 = (1.0)_2 \times 2^{-3}$, etc. A floating point number is one that can be stored *exactly* on a computer.

Single precision and double precision. A single precision number is 32 bits in total, which include 1 bit for the sign, 8 bits for the exponent, and 23 bits for the mantissa. For instance,

$$5.5 = 1.011 \times 2^2 = (\underbrace{0}_{\text{sign}} \,|\, \underbrace{E = 00000010}_{\text{exponent}} \,|\, \underbrace{M = 101100\cdots0}_{\text{mantissa, 23 bits}}). \tag{1}$$

On the other hand, a double precision number is 64 bits in total, which include 1 bit for the sign, 11 bits for the exponent, and 52 bits for the mantissa. Single precision and double precision are standardized by IEEE in the 1980s. In Julia, `x = 7.0` stores a `Float64` type variable, and `x = 7` stores a `Int64` type variable (an 8-byte integer).

To represent negative exponent E, the assumption is that E is offset by 1023, that is,

$$x = \pm M \times 2^{E-1023}. \tag{2}$$

In Julia, `bitstring(x)` returns the IEEE representation of `x`.

1.2 Rounding

Rounding. For instance, $(0.1)_{10} = (1.100\overline{1100})_2 \times 2^{-4}$, so the mantissa must be rounded and its approximation is stored as a floating point number. The dafault rounding strategy is rounding to the nearest. The important thing is the error made in rounding.

$$\text{Absolute rounding error} = |\text{round}(x) - x|, \quad \text{Relative rounding error} = \frac{|\text{round}(x) - x|}{|x|}. \tag{3}$$

IEEE standards. IEEE standard says that it must be the case that

$$\text{round}(a+b) = \text{round}(a) \underbrace{\bigoplus}_{\text{IEEE floating point addition}} \text{round}(b) = (a+b)(1+\delta), \tag{4}$$

where $|\delta| < \epsilon = \textsc{MachEp}$, where $\textsc{MachEp}$ denotes the machine precision. Machine precision is the distance between 1 and the next floating point number. In double precision, that is,

$$\epsilon = -(0|011\cdots1|00\cdots0) + (0|011\cdots1|00\cdots01) = 2^{-52}. \tag{5}$$

IEEE standard also says the following about relative accuracy of computations that

$$\frac{|\text{round}(a+b) - (a+b)|}{|a+b|} = |\delta| < \epsilon. \tag{6}$$

1/30 Lecture

2 Solving Nonlinear Equations

We can find solutions to linear equations such as $3x + 7 = 2$, but we cannot generally find solutions to nonlinear equations such as $\cos x + x^2 - 7 = 5$. This is a root finding problem. A *sufficient* condition for a solution $f(x) = 0$ (f is continuous) to exist is stated in the following theorem.

Theorem 2.1. If f is continuous on $[a, b]$, and if $f(a) \cdot f(b) \leq 0$, then there exists $x \in [a, b]$, such that $f(x) = 0$.

Proof. This is immediate from the Intermediate Value Theorem. □

2.1 Bisection

Idea. We will make use of Theorem 2.1. Split the interval in two, test the condition of a sign change and repeat. Let $a_0 = a$, $b_0 = b$, and from the original interval, let $[a_l, b_l]$ be the interval obtained after l splittings. Then,

$$b_l - a_l = \frac{b_0 - a_0}{2^l} = \frac{L}{2^l}. \tag{7}$$

Let $x_l = \frac{a_l + b_l}{2}$ be the approximation of the solution to $f(x) = 0$ in the lth step. The question is, when do we stop the splittings, and how many steps of this algorithm do we take? If we want $|x_l - x^*| < \epsilon$ where x^* is the true solution $f(x^*) = 0$, then we choose l such that

$$|x_l - x^*| \leq \frac{b_l - a_l}{2} = \frac{L}{2^{l+1}} < \epsilon \implies 2^{l+1} > \frac{L}{\epsilon} \implies l > 1 + \log_2\left(\frac{L}{\epsilon}\right). \tag{8}$$

Convergence behavior. If $e_l = |x_l - x^*|$ represents the error in the lth step, then $e_{l+1} = \frac{1}{2}e_l$, meaning that the absolute error goes down by a factor of 2 in each iteration. Therefore, we can see that the bisection method is not efficient enough.

2.2 Secant Method

Idea. Bisection only used the sign information. If we use the values, we can do better. The idea is to approximate by a line with an initial guess, find the root of the line, update the guess and repeat. The secant line is

$$s(x) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + f(x_0), \tag{9}$$

which has the root

$$x = x_1 - f(x_1) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}. \tag{10}$$

We take this root as x_2 and repeat the process with x_0 substituted by x_1 and x_1 substituted by x_2. Let $f_k = f(x_k)$ for the sake of simplicity. The secant method generates a sequence of approximations to the root of f by

$$x_{k+1} = x_k - f_k \cdot \frac{x_k - x_{k-1}}{f_k - f_{k-1}}. \tag{11}$$

The convergence properties will be revisited later in this course.

2.3 Newton's Method

Idea. In addition to the values of the function, we can make use of both the value and the derivative of a function f at a single point to form an approximation line, find its root, and update and iterate just as in the secant method. The equation of the tangent line is

$$f(x) = f(x_0) + f'(x_0)(x - x_0), \tag{12}$$

which has the root

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}. \tag{13}$$

We take this root as x_1 and repeat the process with x_0 substituted by x_1. The Newton's method generates a sequence of approximations to the root of f by

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{14}$$

Note that (12) is the first-order Taylor approximation of f.

Convergence behavior. Let $e_k = |\xi - x_k|$ be the absolute error, where ξ is the true root of f. Newton's method converges quadratically, meaning that $e_{k+1} = Ae_k^2$ for some constant A. Note that the absolute error is very small, for instance, initially 10^{-1}. Then with only 5 iterations, the error will be near machine precision.

Theorem 2.2 (Theorem 1.8 from Suli and Mayers)**.** Suppose that f is twice continuously differentiable on the interval $I_\delta = [\xi - \delta, \xi + \delta]$, $\delta > 0$, with $f(\xi) = 0$ and $f'(\xi) \neq 0$. Also, assume $A > 0$ such that

$$\left|\frac{f''(x)}{f'(y)}\right| \leq A, \qquad \forall x, y \in I_\delta. \tag{15}$$

If the initial guess satisfies $|\xi - x_0| \leq h$, with $h \leq \min\left(\delta, \frac{1}{A}\right)$, then the sequence $\{x_k\}$ defined by Newton's method (14) converges quadratically to ξ.

2/1 Lecture

Proof. By Taylor's Theorem, we have that

$$f(\xi) = 0 = f(x_0) + f'(x_0)(\xi - x_0) + \frac{f''(\xi_0)}{2}(\xi - x_0)^2, \tag{16}$$

where $\xi_0 \in [\xi, x_0] \subseteq I_\delta$. By Newton's method, we have that

$$\xi - x_1 = \xi - x_0 + \frac{f(x_0)}{f'(x_0)} = -\frac{f''(\xi_0)}{2f'(x_0)}(\xi - x_0)^2. \tag{17}$$

Therefore, we can deduce that

$$|\xi - x_1| \leq \frac{1}{2}\left|\frac{f''(\xi_0)}{f'(x_0)}\right| |\xi - x_0|^2 \leq \frac{1}{2}A\,|\xi - x_0|^2 \leq \frac{1}{2}Ah \cdot \frac{1}{A} = \frac{h}{2}. \tag{18}$$

Repeating this step k times, we can thus show that

$$|\xi - x_k| \leq \frac{h}{2^k} \implies \lim_{k\to\infty} x_k = \xi, \tag{19}$$

proving convergence. Furthermore, since

$$|\xi - x_{k+1}| \leq \frac{1}{2}\left|\frac{f''(\xi_k)}{f'(x_k)}\right| |\xi - x_k|^2, \tag{20}$$

we can see that

$$\lim_{k\to\infty} \frac{|\xi - x_{k+1}|}{|\xi - x_k|^2} = \lim_{k\to\infty}\left|\frac{f''(\xi_k)}{f'(x_k)}\right| = \frac{1}{2}\left|\frac{f''(\xi)}{f'(\xi)}\right| = \texttt{const}. \tag{21}$$

This is exactly the *definition of quadratic convergence.* □

Failures of Newton's method.

- When Newton's method fails to converge (quadratically), it is usually because $f'(\xi) = 0$ at the root. In this case, (15) may not hold, *i.e.*, the quantity may not remain bounded.
- For some initial guesses, Newton's method may fail to converge at all.

Rates of Convergence

For the convergent sequence $e_k = |x_k - x^*|$, we can consider the following limit

$$\lim_{k\to\infty} \frac{e_{k+1}}{e_k^\alpha} = \mu. \tag{22}$$

For instance, Bisection method has $\alpha = 1$ (*i.e.*, linear convergence), Secant method has approximately $\alpha = 1.62$, and Newton's method has $\alpha = 2$ (*i.e.*, quadratic convergence). For $\alpha = 1$, we further define the *order* of convergence as follows: For $\alpha = 1$ and $\mu < 1$, set $\rho = -\log_{10} \mu$ as the *asymptotic rate of convergence.* As an illustration, if $\mu = 1/10$, this means that

$$\frac{e_{k+1}}{e_k} \approx \frac{1}{10}, \tag{23}$$

i.e., the error goes down by 10% every iteration. That is to say that x_k gets one more correct digit every iteration, and we can see that

$$\rho = -\log_{10} \mu = -\log_{10} \frac{1}{10} = 1. \tag{24}$$

3 Solving Nonlinear Systems

A nonlinear system of equations in n variables $x_1, \cdots, x_n$ is given by

$$\begin{aligned} f_1(x_1, \cdots, x_n) &= 0, \\ &\vdots \\ f_n(x_1, \cdots, x_n) &= 0, \end{aligned} \tag{25}$$

which can be condense into the notation

$$\vec{f}(\vec{x}) = \vec{0}, \tag{26}$$

where

$$\vec{f} = \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_n(\vec{x}) \end{pmatrix}, \qquad \vec{x} = \begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix}. \tag{27}$$

Note that:

- Bisection does not, in general, extend to higher dimensions, since sign changes do not necessarily indicate roots.
- The idea behind Secant method and Newton's method directly extends to higher dimensions: Linearize (approximate by a linear function) and find the root of the approximant.

Vector and Matrix Norms

Definition 3.1. Suppose $\vec{u}, \vec{v}, \vec{x} \in \mathbb{C}^n$, then $\|\cdot\|$ is a *norm* on $\mathbb{C}^n$ if

(1) $\|\vec{x}\| \geq 0$, and $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$.

(2) $\|\alpha\vec{x}\| = |\alpha|\|\vec{x}\|$ for $\alpha \in \mathbb{C}$.

(3) $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$ (triangle inequality).

The most useful norm in Numerical Analysis is the l_2-norm, defined as $\|\vec{u}\|_2 = \sqrt{\sum_i |u_i|^2}$. Other norms commonly used include:

- l_∞-norm: $\|\vec{u}\|_\infty = \max_i |u_i|$.
- l_1-norm: $\|\vec{u}\|_1 = \sum_i |u_i|$.
- l_p-norm: $\|\vec{u}\|_p = (\sum_i |u_i|^p)^{1/p}$.

Definition 3.2. Suppose $A, B \in \mathbb{C}^{m \times n}$, then $\|\cdot\|$ is a *matrix norm* if

(1) $\|A\| \geq 0$, and $\|A\| = 0$ if and only if $A = 0$.

(2) $\|\alpha A\| = |\alpha|\|A\|$ for $\alpha \in \mathbb{C}$.

(3) $\|A + B\| \leq \|A\| + \|B\|$.

If $\|\cdot\|$ is any vector norm, then the *induced matrix norm* is defined as

$$\|A\| = \max_{\|\vec{u}\|=1} \|A\vec{u}\| = \max_{\vec{u}} \frac{\|A\vec{u}\|}{\|\vec{u}\|}. \tag{28}$$

2/6 Lecture

For instance, if $\|\vec{u}\| = \|\vec{u}\|_1 = \sum_{i=1}^n |u_i|$, then the induced matrix norm on $A \in \mathbb{C}^{m\times n}$ is

$$\|A\|_1 = \max_{j=1,\cdots,n} \sum_{i=1}^n |A_{ij}|. \tag{29}$$

To see this, note that

$$\|A\vec{u}\|_1 = \left\| \sum_{j=1}^n u_j \vec{A}_j \right\|_1 \leq \sum_{j=1}^n |u_j| \|\vec{A}_j\|_1 \leq \left(\max_j \|\vec{A}_j\|_1 \right) \left(\sum_{j=1}^n |u_j| \right) = \left(\max_j \sum_{i=1}^n |A_{ij}| \right) \|\vec{u}\|_1. \tag{30}$$

Therefore, we have that

$$\frac{\|A\vec{u}\|_1}{\|\vec{u}\|_1} \leq \max_j \sum_{i=1}^n |A_{ij}|. \tag{31}$$

But taking $\vec{u}$ as a zero vector with only the jth entry as 1, we can see that $\|A\vec{u}\|_1 \geq \|\vec{A}_j\|_1$, so the equality must hold for some $\vec{u}$. Therefore, we can conclude that

$$\|A\|_1 = \max_{\vec{u}} \frac{\|A\vec{u}\|}{\|\vec{u}\|} = \max_{j=1,\cdots,n} \sum_{i=1}^n |A_{ij}|. \tag{32}$$

Theorem 3.3. Let $A \in \mathbb{R}^{m\times n}$ be a matrix. Then

$$\|A\|_2 = \sqrt{\max_j \lambda_j}, \tag{33}$$

where λ_j is the jth eigenvalue of $A^\top A$.

Proof. We have that

$$\|A\vec{u}\|^2 = (A\vec{u}, A\vec{u}) = (A\vec{u})^\top A\vec{u} = \vec{u}^\top A^\top A\vec{u}. \tag{34}$$

Note that $A^\top A$ is real and symmetric, it is diagonalizable, so we can write

$$\|A\vec{u}\|^2 = \vec{u}^\top P^\top DP\vec{u}, \tag{35}$$

where D is a diagonal matrix of eigenvalues and P is orthogonal (*i.e.*, $P^\top = P^{-1}$). Therefore, we can compute that

$$\max_{\|\vec{u}\|=1} \|A\vec{u}\|_2^2 = \max_{\|\vec{u}\|=1} \|\vec{u}^\top P^\top DP\vec{u}\|_2^2 = \max_{\|\vec{y}\|=1} \|\vec{y}^\top D\vec{y}\|_2^2 = \max_{\|\vec{y}\|=1} \sum_{i=1}^n y_i^2 \lambda_i = \max_i \lambda_i, \tag{36}$$

where we can change $\vec{y} = P\vec{u}$ without changing the norm since P is diagonal, and the proof of the last equality is left as an *exercise*. $\square$

Condition number

The *condition number* of the problem is the sensitivity of the "problem" at the solution. For instance, consider a function $y = f(x)$, the how sensitive is y to x?

- In an absolute sense, we can write that

$$|y - y'| = C(x)|x - x'|, \tag{37}$$

  where $C(x)$ is the absolute condition number at x. Note that $C(x) \approx |f'(x)|$ when x' is close to x.

- In a relative sense, we can write that

$$\frac{|y' - y|}{|y|} = K(x)\frac{|x' - x|}{|x|}, \tag{38}$$

  where $K(x)$ is the relative condition number at x. Note that $K(x) \approx \left|\frac{xf'(x)}{f(x)}\right|$ when x' is close to x.

Conditioning for matrix equations. Consider solving $A\vec{x} = \vec{b}$, where the input is $\vec{b}$ and the output is $\vec{x} = A^{-1}\vec{b}$. Let $\|\vec{b}' - \vec{b}\|$ be small, and let $\vec{x}' = A^{-1}\vec{b}'$, $\vec{x} = A^{-1}\vec{b}$. Then we have that

$$\|\vec{x}' - \vec{x}\| = \|A^{-1}\vec{b}' - A^{-1}\vec{b}\| \leq \|A^{-1}\| \cdot \|\vec{b}' - \vec{b}\|, \tag{39}$$

where $\|A^{-1}\|$ denotes the induced matrix norm, and in this case the absolute condition number of A at $\vec{b}$. However, recall that the absolute condition number tells us nothing about the number of correct digits in the answer, so we need the relative condition number, which can be computed as

$$\frac{\|\vec{x}' - \vec{x}\|}{\|\vec{x}\|} \leq \|A^{-1}\| \cdot \frac{\|\vec{b}' - \vec{b}\|}{\|\vec{x}\|} = \|A^{-1}\| \cdot \frac{\|\vec{b}' - \vec{b}\|}{\|\vec{b}\|} \cdot \frac{\|\vec{b}\|}{\|\vec{x}\|} = \|A^{-1}\| \cdot \frac{\|\vec{b}' - \vec{b}\|}{\|\vec{b}\|} \cdot \frac{\|A\vec{x}\|}{\|\vec{x}\|} \leq \|A^{-1}\|\|A\| \cdot \frac{\|\vec{b}' - \vec{b}\|}{\|\vec{b}\|}, \tag{40}$$

so that $\|A^{-1}\|\|A\|$ is the relative condition number of A at $\vec{b}$.

The Singular Value Decomposition

If $A \in \mathbb{R}^{m\times n}$ with $m \geq n$, then we can write

$$A = U\Sigma V^{\top}, \tag{41}$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are diagonal, and $\Sigma \in \mathbb{R}^{m\times n}$ is a rectangular diagonal matrix.

2/8 Lecture

Recall that the Taylor series of a scalar-valued function of several variables is

$$f = f(x_1, \cdots, x_n) = f(\vec{y}) + \sum_j \frac{\partial f}{\partial x_j}(x_j - y_j) + \textit{higher order terms}. \tag{42}$$

3.1 Multivariable Vector-Valued Taylor Series

Consider the multivariable vector-valued function

$$\vec{f}(\vec{x}) = \begin{pmatrix} f_1(x_1, \cdots, x_m) \\ \vdots \\ f_n(x_1, \cdots, x_m) \end{pmatrix}. \tag{43}$$

Place the expansions back into $\vec{f}$ and collect the terms, we have that

$$\vec{f}(\vec{x}) = \begin{pmatrix} f_1(\vec{y}) + \sum_{k=1}^{n} \frac{\partial f_1(\vec{y})}{\partial x_k}(x_k - y_k) + \cdots \\ \vdots \\ f_n(\vec{y}) + \sum_{k=1}^{n} \frac{\partial f_n(\vec{y})}{\partial x_k}(x_k - y_k) + \cdots \end{pmatrix} = \vec{f}(\vec{y}) + \begin{pmatrix} \frac{\partial f_1(\vec{y})}{\partial x_1} & \cdots & \frac{\partial f_1(\vec{y})}{\partial x_m} \\ \vdots & & \vdots \\ \frac{\partial f_n(\vec{y})}{\partial x_1} & \cdots & \frac{\partial f_n(\vec{y})}{\partial x_m} \end{pmatrix} \begin{pmatrix} x_1 - y_1 \\ \vdots \\ x_m - y_m \end{pmatrix}$$

$$= \vec{f}(\vec{y}) + J(\vec{y})(\vec{x} - \vec{y}) + (\vec{x} - \vec{y})^\top \cdot \frac{Q(\vec{y})}{2} \cdot (\vec{x} - \vec{y}) + \textit{higher order terms}, \tag{44}$$

where J is called the *Jacobian matrix*, and Q is a *tensor* defined so that the ith entry of the second-order term $(\vec{x} - \vec{y})^\top \cdot \frac{Q(\vec{y})}{2} \cdot (\vec{x} - \vec{y})$ is given by

$$\frac{1}{2} \sum_{k,l} \frac{\partial^2 f_i(\vec{y})}{\partial x_k \partial x_l}(x_k - y_k)(x_l - y_l) = \frac{1}{2}(\vec{x} - \vec{y})^\top H_i(\vec{y})(\vec{x} - \vec{y}), \tag{45}$$

and H_i is the *Hessian* for f_i.

3.2 Multivariate Newton's Method

We want to solve a square system $\vec{f}(\vec{x}) = \vec{0}$ (*i.e.*, $m = n$). In general, determining whether there is one, none, or multiple solutions is exceedingly difficult. However, if a approximate $\vec{f}$ at $\vec{x}_0$ near the root $\vec{\xi}$, then we have

$$\vec{f}(\vec{x}) \approx \vec{f}(\vec{x}_0) + J(\vec{x}_0)(\vec{x} - \vec{x}_0), \tag{46}$$

also known as the truncated multivariable Taylor series. Evaluting at $\vec{xi}$, we have that

$$\vec{f}(\vec{x}_i) = \vec{0} \approx \vec{f}(\vec{x}_0) + J(\vec{x}_0)(\vec{\xi} - \vec{x}_0) \implies -\vec{f}(\vec{x}_0) \approx J(\vec{x}_0)(\vec{\xi} - \vec{x}_0) \implies \vec{\xi} \approx \vec{x}_0 - J^{-1}(\vec{x}_0)\vec{f}(\vec{x}_0). \tag{47}$$

Therefore, the Newton iteration is

$$\vec{x}_{k+1} = \vec{x}_k - J^{-1}(\vec{x}_k)\vec{f}(\vec{x}_k). \tag{48}$$

It can be shown that convergence is also quadratic, meaning that

$$\|\vec{x}_{k+1} - \vec{\xi}\|_2 \approx \|\vec{x}_k - \vec{\xi}\|_2^2. \tag{49}$$

Much like Newton's method, quadratic convergence may suffer if we have a singular Jacobian $J(\vec{x}_k)$. Another commonly seen form of the multivariate Newton's method is

$$J(\vec{x}_k)(\vec{x}_{k+1} - \vec{x}_k) = f(\vec{x}_k), \tag{50}$$

which can be done by solving $J(\vec{x}_k)\vec{s}_k = f(\vec{x}_k)$ and setting $\vec{x}_{k+1} = \vec{x}_k + \vec{s}_k$ in the iterative step. We will revise this when we talk about optimization.

2/13 Lecture

4 Optimization

4.1 Single Variable Optimization

Recall that root finding solves $f(x) = 0$. If we instead want to maximize or minimize f, we want to look for x such that $f'(x) = 0$. Therefore, we can apply our root finding methods to the equation $f'(x) = 0$ instead. Newton's method then becomes

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}. \tag{51}$$

However, note that for a general continuous function, $f'(x) = 0$ does not necessarily imply a global optimum. It can imply local optimums, or even non-optimal points (for instance, $x = 0$ for $f(x) = x^3$). One (common) regime where we can guarantee a minimum or maximum is when f is convex, defined as follows.

Definition 4.1. f is *convex* on $[a, b]$ if

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y), \tag{52}$$

for all $0 \leq \alpha \leq 1$ and $x, y \in [a, b]$.

Theorem 4.2. If f is convex on $[a, b]$, then any local minimum of f is also a global minimum of f.

Proof. We prove by contradiction. Let x^* be a local minimum of f, and $\tilde{x} \neq x^*$ be such that $f(\tilde{x}) < f(x^*)$. Then, since f is convex, we have that

$$f(\alpha x^* + (1-\alpha)\tilde{x}) \leq \alpha f(x^*) + (1-\alpha)f(\tilde{x}) < \alpha f(x^*) + (1-\alpha)f(x^*) = f(x^*). \tag{53}$$

But since f is convex and x^* is a local minimum of f, there exists some small neighborhood $B_\epsilon(x^*)$ around x^*, such that $f(x) \geq f(x^*)$ for all $x \in B_\epsilon(x^*)$. Taking $\alpha = 1 - \frac{\epsilon}{2|x^* - \tilde{x}|}$ (ϵ can be arbitrarily small so we can guarantee that $0 \leq \alpha \leq 1$), we observe that

$$|\alpha x^* + (1-\alpha)\tilde{x} - x^*| = (1-\alpha)|x^* - \tilde{x}| = \frac{\epsilon}{2} < \epsilon, \tag{54}$$

so $\alpha x^* + (1-\alpha)\tilde{x} \in B_\epsilon(x^*)$. But (53) then leads to a contradiction, so the proof is complete. □

Theorem 4.3. If f is *strictly convex*, then there is only one local minimum.

Proof. Strictly convexity requires the inequality in (52) to be strict. The proof is trivial. □

4.2 Multivariate Optimization

Multivariate Newton's method. Now consider that we want to find

$$\vec{x}^* = \underset{\vec{x} \in \mathscr{X}}{\arg\min}\, f(\vec{x}) = \underset{\vec{x} \in \mathscr{X}}{\arg\min}\, f(x_1, x_2, \cdots, x_n). \tag{55}$$

Assume that f is smooth and strictly convex, then we can (hopefully) solve this using Newton's method on the system $\vec{\nabla} f = \vec{0}$, which is equivalent to $\partial f / \partial x_i = 0$ for all $i = 1, \cdots, n$. Then Newton's method says that

$$\begin{aligned}
\vec{x}_{k+1} &= \vec{x}_k - \begin{pmatrix} \frac{\partial}{\partial x_1}\left(\frac{\partial f(\vec{x}_k)}{\partial x_1}\right) & \cdots & \frac{\partial}{\partial x_n}\left(\frac{\partial f(\vec{x}_k)}{\partial x_1}\right) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1}\left(\frac{\partial f(\vec{x}_k)}{\partial x_n}\right) & \cdots & \frac{\partial}{\partial x_n}\left(\frac{\partial f(\vec{x}_k)}{\partial x_n}\right) \end{pmatrix}^{-1} \vec{\nabla} f(\vec{x}_k) \\
&= \vec{x}_k - \begin{pmatrix} \frac{\partial^2 f(\vec{x}_k)}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\vec{x}_k)}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\vec{x}_k)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\vec{x}_k)}{\partial x_n \partial x_n} \end{pmatrix}^{-1} \vec{\nabla} f(\vec{x}_k) = \vec{x}_k - H^{-1}(\vec{x}_k)\vec{\nabla} f(\vec{x}_k),
\end{aligned} \tag{56}$$

where H denotes the Hessian matrix. There is one computational challenge with this: evaluating and inverting the Hessian matrix are expensive, taking $O(n^2/2)$ and $O(n^3)$ time respectively. To avoid explicit construction of H, we consider *Quasi-Newton methods*, and recall that the secant method is the most basic Quasi-Newton method.

Broyden's Update. The goal is to form an approximation to H on each step of the multivariate Newton's method for optimization, and to be able to cheaply update this approximation. For instance, we want something like

$$\vec{x}_1 = \vec{x}_0 - H^{-1}(\vec{x}_0)\vec{\nabla} f(\vec{x}_0), \quad \vec{x}_2 = \vec{x}_1 - H^{-1}(\vec{x}_1)\vec{\nabla} f(\vec{x}_1), \quad H(\vec{x}_1) = H(\vec{x}_0) + \textit{something}. \tag{57}$$

Now consider the implied linear approximation from the multivariate Newton's method for optimization, that is,

$$\vec{\nabla} f(\vec{x}_{k+1}) \approx \vec{0} \approx \underbrace{\vec{\nabla} f(\vec{x}_k)}_{=:\vec{\nabla} f_k} + \underbrace{H(\vec{x}_k)}_{:=H_k} \underbrace{(\vec{x}_{k+1} - \vec{x}_k)}_{=:\vec{s}_k} \implies H_k \vec{s}_k \approx -\vec{\nabla} f_k. \tag{58}$$

To continue the iteration, we want to update the approximate Hessian matrix to H_{k+1}. Similar to the secant method, it should also be the case that

$$\vec{\nabla} f_{k+1} - \vec{\nabla} f_k \approx H_{k+1}(\vec{x}_{k+1} - \vec{x}_k) \implies H_{k+1}\vec{s}_k \approx \vec{\nabla} f_{k+1} - \vec{\nabla} f_k. \tag{59}$$

Of course this does not uniquely determine H_{k+1} given $\vec{s}_k$, $\vec{\nabla} f_{k+1}$, and $\vec{\nabla} f_k$ are known. We need an extra constraint that $H_{k+1} - H_k$ is of rank 1. This gives that

$$H_{k+1} = H_k + \frac{1}{\vec{s}_k^\top \vec{s}_k}\left(\vec{\nabla} f_{k+1} - \vec{\nabla} f_k - H_k \vec{s}_k\right)\vec{s}_k^\top. \tag{60}$$

Note that the form $\vec{u}\vec{v}^\top$ is the outer product $\vec{u} \otimes \vec{v}$, while the form $\vec{u}^\top \vec{v}$ is the inner product $\vec{u} \cdot \vec{v}$. Therefore, we propose the corresponding algorithm as follows. First, we initialize H_0 (finite differences, diagonal, etc.). Then, we set $\vec{x}_1 = \vec{x}_0 - H_0^{-1}\vec{\nabla} f(\vec{x}_0)$ and update to H_1 based on (60). Iterate this step until the terminating condition is met.

Remark 4.4. Since we are interested only in H^{-1}, there is actually a better way to update H_{k+1}^{-1} given H_k^{-1} without referring to H_{k+1} or H_k. Note that the updated Hessian matrix is in the form

$$H_{k+1} = H_k + \vec{u}\vec{v}^\top. \tag{61}$$

Therefore, we can use the *Sherman-Morrison-Woodbury formula*, which states that

$$(A + \vec{u}\vec{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\vec{u}\vec{v}^\top A^{-1}}{1 + \vec{v}^\top A^{-1}\vec{u}}. \tag{62}$$

2/15 Lecture

5 Numerical Linear Algebra

In infinite precision-exact arithmetic, solving linear systems comes down to a question of whether the system is consistent or not. On the computer, however, the situation is different due to the round-off error and finite precision calculations. We first make a few remarks on Julia. In Julia, the jth column of a matrix can be accessed by `A[:,j]`. A submatrix can be extracted as `A[i1:i2,j1:j2]`. To generate a matrix, one can do something like `A=rand(5,4)`. To compute the induced p-norm, one can use `opnorm(A,p)`. Note that `p=Inf` is also an option.

5.1 Gaussian Elimination

We all know how Gaussian elimination works. Consider a 2×2 matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \tag{63}$$

Then $A\vec{x} = \vec{b}$ can be solved by eliminating a_{21}, such that

$$\left(\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{array}\right) \sim \left(\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ 0 & a_{22} - \frac{a_{12}a_{21}}{a_{11}} & b_2 - \frac{a_{12}a_{21}}{a_{11}} \end{array}\right). \tag{64}$$

This implies that

$$x_2 = \frac{b_2 - \frac{a_{12}a_{21}}{a_{11}}}{a_{22} - \frac{a_{12}a_{21}}{a_{11}}}, \qquad x_1 = \frac{b_1 - a_{12}x_2}{a_{11}}. \tag{65}$$

This process is called the *backward substitution*, but it can easily fail. For instance, if one or some of the denominators evaluate to zero, then the process can no longer proceed. A successful algorithm would need to involve pivoting rows, and we will return to this later. One question to consider for this moment is how expensive is Gaussian elimination. To see this, we need to count the number of floating-point operations required to put A in echelon form. Consider the following algorithm:

```
for columns j = 1, ··· , n − 1 do
    for rows i = j + 1, ··· , n do
        Compute val = a_ij / a_jj;
        Compute row_i − val · row_j;
        Compute b_i − val · b_j;
    end for
end for
```

Within each inner loop, the first computation takes 1 operation, the second computation takes $2(n-j)$ computations, and the third operation takes 2 flops. Therefore, the total number of floating-point operations can be computed as

$$\sum_{j=1}^{n-1}\sum_{i=j+1}^{n}(2(n-j)+3) = \sum_{j=1}^{n-1}(n-j)(2(n-j)+3) = \Theta(n^3). \tag{66}$$

5.2 LU Factorization

Another way to think about Gaussian elimination is the LU factorization. Each row operation corresponds to a matrix multiplication by an *elementary* lower triangular matrix on the left. For instance, consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}. \tag{67}$$

We can perform some elementary row operations on A such that

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ -7 & 0 & 1 \end{pmatrix}}_{L_1}\underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}}_{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -21 \end{pmatrix}, \tag{68}$$

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}}_{L_2}\underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -21 \end{pmatrix}}_{L_1A} = \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & -9 \end{pmatrix}}_{U}. \tag{69}$$

Therefore, we can see that $L_2L_1A = U$, which implies $A = L_1^{-1}L_2^{-1}U$. Note that $L_1^{-1}L_2^{-1}$ is also lower triangular, then solving $A\vec{x} = \vec{b}$ is equivalent to solving $LU\vec{x} = \vec{b}$, where L is a lower triangular matrix and U is an upper triangular matrix. To do this, we only need to solve $L\vec{y} = \vec{b}$ by forward substitution, and then solve $U\vec{x} = \vec{y}$ using backward substitution. Each of these steps take $O(n^2)$ operations. However, the LU factorization step requires the same number of operations as the Gaussian elimination, which is $O(n^3)$, thus dominating the runtime.

Gaussian elimination and LU factorization failure. We have to notice that Gaussian elimination and LU factorization may fail if there is a zero or a small number in the pivot position. The solution is to interchange some rows, resulting in a factorization of the form $PA = LU$, where P is a permutation (row exchanging) matrix. For instance, let $\epsilon <$ MachEp, then consider

$$\begin{pmatrix} \epsilon & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}. \tag{70}$$

The exact solution to this linear system is

$$x_1 = 1 + \frac{\epsilon}{\epsilon - 1}, \qquad x_2 = 1 - \frac{\epsilon}{\epsilon + 1}, \tag{71}$$

and note that x_1 and x_2 are both approximately 1. Without pivoting, in floating point arithmetic, Gaussian elimination will give us

$$\left(\begin{array}{cc|c} \epsilon & 1 & 1 \\ 1 & 1 & 2 \end{array}\right) \sim \left(\begin{array}{cc|c} \epsilon & 1 & 1 \\ 0 & 1-1/\epsilon & 2-1/\epsilon \end{array}\right) \approx \left(\begin{array}{cc|c} \epsilon & 1 & 1 \\ 0 & -1/\epsilon & -1/\epsilon \end{array}\right), \tag{72}$$

which would further result in

$$x_2 = 1, \qquad x_1 = \frac{1-1}{\epsilon} = 0, \tag{73}$$

throwing the wrong answer.

2/22 Lecture

Before discussing pivoted LU factorization, there is one special case where it can be shown that pivoting is not necessary: $\vec{x}^\top A\vec{x} > 0$, *i.e.*, A is a *symmetric positive definite* matrix. Note that this fact is one of the earliest results in numerical analysis.

5.3 Cholesky Factorization

If A is a symmetric positive definite matrix, then we can write it as $A = U^\top U$, since then $A^\top = U^\top U$ as well. The algorithm is straightforward. Set

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}, \tag{74}$$

where $u_{jj} \neq 1$ for all j (in fact, this condition is not necessary). Then, we have that

$$U^\top U = \begin{pmatrix} u_{11} & 0 & \cdots & 0 \\ u_{12} & u_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ u_{1n} & u_{2n} & \cdots & u_{nn} \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix} = \begin{pmatrix} u_{11}^2 & u_{11}u_{12} & \cdots & u_{11}u_{1n} \\ u_{12}u_{11} & \sum_{i=1}^2 u_{i2}^2 & \cdots & \sum_{i=1}^2 u_{i2}u_{in} \\ \vdots & \vdots & \ddots & \vdots \\ u_{1n}u_{11} & \sum_{i=1}^2 u_{in}u_{i2} & \cdots & \sum_{i=1}^n u_{in}^2 \end{pmatrix}. \tag{75}$$

Therefore, if we set $A = U^\top U$, we can iteratively compute $u_{11} = \sqrt{a_{11}}$, $u_{12} = a_{12}/u_{11}$, and so on. The total cost would be $\Theta(n^3/3)$ for the Cholesky factorization algorithm.

5.4 Row Pivoting

When pivoting is necessary when doing an LU decomposition, we can effectively write

$$L_m P_{m-1} \cdots L_3 P_2 L_2 P_1 L_1 A = U, \tag{76}$$

where each L_j is a lower triangular matrix and each P_j is a permutation matrix. By the term "permutation matrix", we refer to some matrix P_0 that can for instance interchange two rows, such that

$$P_0 A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} - & \vec{a}_1^\top & - \\ - & \vec{a}_2^\top & - \\ - & \vec{a}_3^\top & - \\ - & \vec{a}_4^\top & - \end{pmatrix} = \begin{pmatrix} - & \vec{a}_1^\top & - \\ - & \vec{a}_2^\top & - \\ - & \vec{a}_4^\top & - \\ - & \vec{a}_3^\top & - \end{pmatrix}. \tag{77}$$

The question is, how to obtain an equivalent form of the factorization $PA = LU$, where P is the permutation matrix resulting from all of the row swaps in the factorization (possibly reordering all rows), L is a lower triangular matrix, and U is an upper triangular matrix. Moreover, note that since each P_j involves only a single row interchange (to be distinguished from P), we have that $P_j^{-1} = P_j = P_j^\top$. As an example, suppose that $L_2 P_1 L_1 A = U$. We then take $L_2 = L_2'$ and $L_1 = P_1 L_1' P_1$. This thus implies that

$$L_2' P_1 P_1 L_1' P_1 A = U \implies L_2' L_1' P_1 A = U \implies P_1 A = (L_2' L_1')^{-1} U. \tag{78}$$

It can be easily shown that $L_1' = P_1 L_1 P_1$ and $L_2' = L_2$ are lower triangular, and therefore in the general case, all permutation matrices can be "moved" to the right by the above argument (details omitted, but make heavy use of the fact that $P_j = P_j^{-1}$ as above).

Gaussian elimination and LU factorization failure (revisited). Recall a previous example, in which we set $\epsilon < \text{MachEp}$ and consider

$$\begin{pmatrix} \epsilon & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}. \tag{79}$$

The exact solution to this linear system is

$$x_1 = 1 + \frac{\epsilon}{\epsilon - 1}, \qquad x_2 = 1 - \frac{\epsilon}{\epsilon + 1}, \tag{80}$$

but Gaussian elimination without pivoting would result in

$$x_2 = 1, \qquad x_1 = \frac{1-1}{\epsilon} = 0. \tag{81}$$

Now with row pivoting, we swap the first and second rows, so that

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ \epsilon & 1 & 1 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1-\epsilon & 1-2\epsilon \end{array}\right) \approx \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array}\right), \tag{82}$$

resulting in the correct answer

$$x_2 = 1, \qquad x_1 = \frac{2-1}{1} = 1. \tag{83}$$

5.5 Conditioning and Backward Stability

Recall from before that we computed the (relative) condition number of solving $A\vec{x} = \vec{b}$, which is

$$K(A) = \|A\|\|A^{-1}\|. \tag{84}$$

Theorem 5.1 (Perturbations of $A, \vec{x}$)**.** If $A(\vec{x} + \Delta\vec{x}) = \vec{b} + \Delta\vec{b}$, then

$$\frac{\|\Delta\vec{x}\|}{\|\vec{x}\|} \leq K(A)\frac{\|\Delta\vec{b}\|}{\|\vec{b}\|}. \tag{85}$$

If $(A + \Delta A)(\vec{x} + \Delta\vec{x}) = \vec{b}$, then

$$\frac{\|\Delta\vec{x}\|}{\|\vec{x}\|} \leq K(A)\frac{\|\Delta A\|}{\|A\|}, \qquad \text{as } \|\Delta A\| \to 0. \tag{86}$$

Note that for any induced matrix norm, we have that

$$1 = \|I\| = \|AA^{-1}\| \leq \|A\|\|A^{-1}\| = K(A), \tag{87}$$

so a condition number of 1 is the best we can hope for. In that case, the relative perturbation of the solution has the same size as that of the data. A condition number of size 10^t indicates that in floating point arithmetic, roughly t digits are lost (*i.e.*, become incorrect) in computing the solution $\vec{x}$. Moreover, if $K(A) > \text{MachEp}^{-1}$, then for computational purposes the matrix A is singular.

Residual and backward error. Suppose that $\tilde{x}$ is the computed solution to the exact linear system $A\vec{x} = \vec{b}$, then $\tilde{x} = \vec{x}$ is unknown since the exact solution $\vec{x}$ is unknown. Define the *residual* as

$$\vec{r} = \vec{b} - A\tilde{x}, \tag{88}$$

and note that $\tilde{x}$ solves the linear system $A\tilde{x} = \vec{b} - \vec{r}$, a perturbed linear system. The *backward error* is then $\|\vec{r}\|$, which the the perturbation from the original problem to the one that is solved exactly. However, note that small backward error does not necessarily imply small $\|\tilde{x} - \vec{x}\|$. To see this, let $\vec{h} = \tilde{x} - \vec{x}$ and thus $A\vec{h} = -\vec{r}$. By Theorem 5.1, we have that

$$\frac{\|\vec{x} - \tilde{x}\|}{\|\vec{x}\|} \leq K(A)\frac{\|r\|}{\|b\|}. \tag{89}$$

This says that the relative error can be much larger than the relative residual when the condition number $K(A)$ is large. In other words, when solving a linear system, all that can be expected is that the backward error be small, but we cannot guarantee a bound on the error.

5.6 QR Factorization

For the moment, let $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $\text{rank}(A) = n$ (*i.e.*, n linearly independent columns). For a general $\vec{b} \in \mathbb{R}^m$, the system $A\vec{x} = \vec{b}$ has no solution if $m > n$. We can, however, ask for the least-squares solution

$$\vec{x}^* = \arg\min_{\vec{x}} \|A\vec{x} - \vec{b}\|_2. \tag{90}$$

In order to achieve this, we require that $\vec{x}^*$ satisfies the *normal equations*

$$A^\top(A\vec{x} - \vec{b}) = \vec{0} \implies A^\top A\vec{x} = A^\top\vec{b}. \tag{91}$$

To see this, let $\vec{y} \in \mathbb{R}^n$ be any vector, then

$$\begin{aligned}\|A(\vec{x}+\vec{y}) - \vec{b}\|_2^2 &= (A\vec{x} + A\vec{y} - \vec{b})^\top(A\vec{x} + A\vec{y} - \vec{b}) = (A\vec{x} - \vec{b})^\top(A\vec{x} - \vec{b}) + 2(A\vec{y})^\top(A\vec{x} - \vec{b}) + (A\vec{y})^\top(A\vec{y}) \\ &= \|A\vec{x} - \vec{b}\|_2^2 + 2y^\top A^\top(A\vec{x} - \vec{b}) + \|A\vec{y}\|_2^2 = \|A\vec{x} - \vec{b}\|_2^2 + \|A\vec{y}\|_2^2 \geq \|A\vec{x} - \vec{b}\|_2^2,\end{aligned} \tag{92}$$

which completes the proof. Furthermore, if we can write $A = QR$ with $Q \in \mathbb{R}^{m\times n}$ orthogonal (*i.e.*, $Q^\top Q = I$) and $R \in \mathbb{R}^{n\times n}$ upper triangular, the normal equations would become

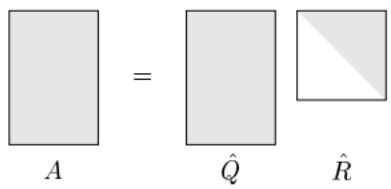
$$R^\top Q^\top QR\vec{x} = R^\top Q^\top\vec{b} \implies R^\top R\vec{x} = R^\top Q^\top\vec{b}. \tag{93}$$

Clearly $R^\top$ is non-singular, so this is equivalent to solving $R\vec{x} = Q^\top\vec{b}$. This is an $n \times n$ square triangular linear system that can be easily solved by backward substitution, and has a solution $\vec{x} = R^{-1}Q^\top\vec{b}$, which is the desired least-squares solution.

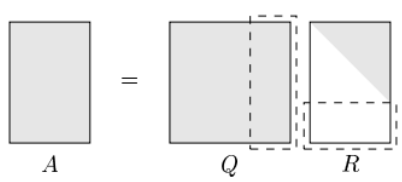
2/27 Lecture

There are in general two scenarios of QR factorization in the case $m \geq n$ and full rank n.

(I) Reduced QR factorization:

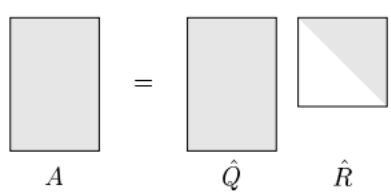


(II) Full QR factorization:



5.7 Gram-Schmidt (Triangular) Orthogonalization

In this section, consider the reduced QR factorization, *i.e.*, factorization of Type (I).



The simple (and naive) way to compute Q is via the *Gram-Schmidt process* which creates an orthonormal basis for the span of the columns of A, *i.e.*, $\vec{a}_1, \cdots, \vec{a}_n \in \mathbb{R}^m$. Write

$$A = \begin{pmatrix} | & | & & | \\ \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \\ | & | & & | \end{pmatrix}, \qquad Q = \begin{pmatrix} | & | & & | \\ \vec{q}_1 & \vec{q}_2 & \cdots & \vec{q}_n \\ | & | & & | \end{pmatrix} \tag{94}$$

If $A = QR$, then since R is an upper triangular matrix, we have that

$$\begin{cases} \vec{a}_1 = & r_{11}\vec{q}_1, \\ \vec{a}_2 = & r_{12}\vec{q}_1 + r_{22}\vec{q}_2, \\ & \vdots \\ \vec{a}_n = & r_{1n}\vec{q}_1 + \cdots + r_{nn}\vec{q}_n. \end{cases} \tag{95}$$

The Gram-Schmidt process says: turn $\vec{a}_1, \cdots, \vec{a}_n$ into $\vec{q}_1, \cdots, \vec{q}_n$, so that Q can be a matrix with orthonormal columns (*i.e.*, an orthogonal matrix). The algorithm is then described as follows: on the jth step, set

$$\vec{v}_j = \vec{a}_j - (\vec{q}_1^\top \vec{a}_j)\vec{q}_1 - \cdots - (\vec{q}_{j-1}^\top \vec{a}_j)\vec{q}_{j-1}, \qquad \vec{q}_j = \frac{\vec{v}_j}{\|\vec{v}_j\|}. \tag{96}$$

Comparing with (95), this implies that

$$r_{ij} = \vec{q}_i^{\,\top} \vec{a}_j, \qquad |r_{jj}| = \left\| \vec{a}_j - \sum_{i=1}^{j-1} r_{ij}\vec{q}_i \right\|_2, \tag{97}$$

since we have that

$$r_{jj}\vec{q}_j = \vec{a}_j - r_{ij}\vec{q}_1 - \cdots - r_{j-1,j}\vec{q}_{j-1}. \tag{98}$$

The pseudocode of this classical Gram-Schmidt orthogonalization algorithm is shown as follows.

```
for j = 1 to n do
    v_j ← a_j;
    for i = 1 to j − 1 do
        r_ij ← q_i^⊤ a_j;
        v_j ← v_j − r_ij q_i;      // round-off error can accumulate
    end for
    r_jj ← ‖v_j‖_2;
    q_j ← v_j / r_jj;
end for
```

Unfortunately, this algorithm is numerically unstable. The updating step in the first equation of (96) will require an iteration over $i = 1, \cdots, j-1$, potentially causing the the round-off error to accumulate. An alternative way to think about the Gram-Schmidt orthogonalization algorithm is as follows. Define

$$Q_{j-1} = \begin{pmatrix} | & | & & | \\ \vec{q}_1 & \vec{q}_2 & \cdots & \vec{q}_{j-1} \\ | & | & & | \end{pmatrix}, \tag{99}$$

and let P_j be the projector onto the subspace orthogonal to the columns of Q_{j-1}, that is,

$$P_j = I - Q_{j-1}Q_{j-1}^\top. \tag{100}$$

The classical Gram-Schmidt orthogonalization algorithm can then be written as

$$\vec{q}_1 = \frac{\vec{a}_1}{\|\vec{a}_1\|} = \frac{P_1\vec{a}_1}{\|P_1\vec{a}_1\|}, \qquad \cdots, \qquad \vec{q}_j = \frac{P_j\vec{a}_j}{\|P_j\vec{a}_j\|}. \tag{101}$$

This modified Gram-Schmidt algorithm is stable, and implicitly writes the projection in a different form

$$P_j = I - Q_{j-1}Q_{j-1}^\top = \underbrace{(I - \vec{q}_{j-1}\vec{q}_{j-1}^\top)}_{P_{\perp \vec{q}_{j-1}}}\underbrace{(I - \vec{q}_{j-2}\vec{q}_{j-2}^\top)}_{P_{\perp \vec{q}_{j-2}}}\cdots\underbrace{(I - \vec{q}_1\vec{q}_1^\top)}_{P_{\perp \vec{q}_1}}. \tag{102}$$

This is mathematically equivalent to (96), but numerically different. The pseudocode of this modified Gram-Schmidt orthogonalization algorithm is shown as follows.

```
for i = 1 to n do
    v_i ← a_i;        // Θ(m)
end for
for i = 1 to n do
    r_ii ← ||v_i||_2;      // m multiplications + m-1 additions + 1 square root
    q_i ← v_i / r_ii;      // m divisions
    for j = i + 1 to n do
        r_ij ← q_i^T v_j;      // m multiplications + m-1 additions
        v_j ← v_j − r_ij q_i;      // m multiplications + m subtractions
    end for
end for
```

This modified Gram-Schmidt algorithm is the one used in practice, since it is less sensitive to the effects of rounding errors, and thus is a stable algorithm. The number of floating-point operations it need can be computed as

$$\sum_{i=1}^{n}\left(3m+\sum_{j=i+1}^{n}(4m-1)\right)\sim\sum_{i=1}^{n}\sum_{j=i+1}^{n}4m\sim\Theta(2mn^2). \tag{103}$$

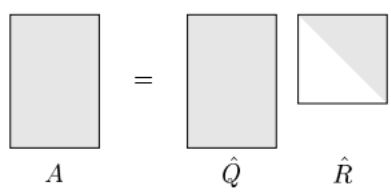
Therefore, this is a cubic algorithm (like Gaussian elimination, basically), and it is suited to reduced QR factorization, *i.e.*, factorization of Type (I).

5.8 Householder Reflection (Orthogonal) Triangularization

As we have seen previously, the Gram-Schmidt iteration applies a succession of elementary triangular matrices R_k on the right of A, so that the resulting matrix

$$A\underbrace{R_1R_2\cdots R_n}_{\hat{R}^{-1}}=\hat{Q} \tag{104}$$

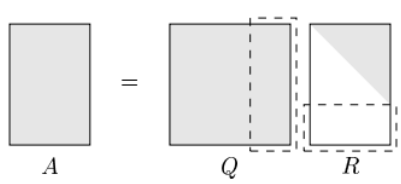
has orthonormal columns, *i.e.*, is an orthogonal matrix. The product $\hat{R}=R_n^{-1}\cdots R_2^{-1}R_1^{-1}$ is an upper triangular matrix, and thus $A=\hat{Q}\hat{R}$ is a reduced QR factorization of A, *i.e.*, a factorization of Type (I).



In contrast, the *Household method* that we are going to introduce applies a succession of elementary matrices Q_k on the left of A, so that the resulting matrix

$$\underbrace{Q_n\cdots Q_2Q_1}_{Q^\top}A=R \tag{105}$$

is an upper triangular matrix. The product $Q=Q_1^\top Q_2^\top\cdots Q_n^\top$ is orthogonal, and thus $A=QR$ would be a full QR factorization of A, *i.e.*, a factorization of Type (II).



As follows is an example of the Household method.

$$\begin{pmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \end{pmatrix} \xrightarrow{Q_1} \begin{pmatrix} \star & \star & \star \\ 0 & \star & \star \\ 0 & \star & \star \\ 0 & \star & \star \\ 0 & \star & \star \end{pmatrix} \xrightarrow{Q_2} \begin{pmatrix} \times & \times & \times \\ & \star & \star \\ & 0 & \star \\ & 0 & \star \\ & 0 & \star \end{pmatrix} \xrightarrow{Q_3} \begin{pmatrix} \times & \times & \times \\ & \times & \times \\ & & \star \\ & & 0 \\ & & 0 \end{pmatrix}. \tag{106}$$

The question is, how can we construct orthogonal matrices Q_k, so as to introduce zeros as indicated above? The standard approach is as follows. Each Q_k is chosen to be a matrix of the form

$$Q_k = \begin{pmatrix} I_{k-1} & 0 \\ 0 & F \end{pmatrix}, \tag{107}$$

where I_{k-1} is the $(k-1)\times(k-1)$ identity matrix and F is an $(m-k+1)\times(m-k+1)$ orthogonal matrix. Multiplication by F must introduce zeros in the kth column. The Householder algorithm chooses F to be a particular matrix called a *Household reflector*. Suppose that at the beginning of Step k, the entries $k, \cdots, m$ of the kth column are given by $\vec{x} \in \mathbb{R}^{m-k+1}$. To introduce the correct zeros into the kth column, the Household reflector F should effect the following map, such that

$$\vec{x} = \begin{pmatrix} \times \\ \times \\ \times \\ \vdots \\ \times \end{pmatrix} \quad \xrightarrow{F} \quad F\vec{x} = \begin{pmatrix} \|\vec{x}\| \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \|\vec{x}\|\vec{e}_1. \tag{108}$$

The idea is that, the reflector F should reflect the space $\mathbb{R}^{m-k+1}$ across the hyperplane H orthogonal to $\vec{v} = \|\vec{x}\|\vec{e}_1 - \vec{x}$, as is demonstrated in Figure 1.

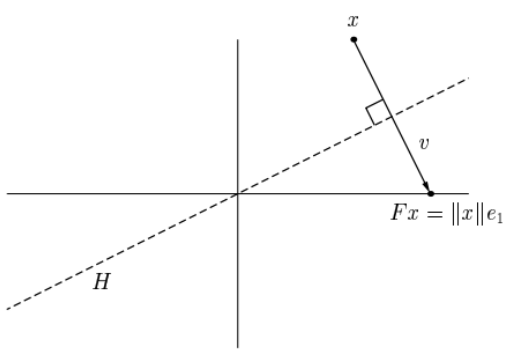


Figure 1: An illustration of a Household reflector.

Recall that for any $\vec{y} \in \mathbb{R}^p$, the *orthogonal projection* of $\vec{y}$ onto a hyperplane H that is orthogonal to some vector $\vec{v} \in \mathbb{R}^p$ can be written as

$$P\vec{y} = \left(I - \frac{\vec{v}\vec{v}^\top}{\vec{v}^\top\vec{v}}\right)\vec{y} = \vec{y} - \vec{v}\left(\frac{\vec{v}^\top\vec{y}}{\vec{v}^\top\vec{v}}\right). \tag{109}$$

Since a reflection would need to go twice the distance as a projection, the reflection would be written as

$$F\vec{y} = \vec{y} - 2\vec{v}\left(\frac{\vec{v}^\top\vec{y}}{\vec{v}^\top\vec{v}}\right) = \left(I - 2\frac{\vec{v}\vec{v}^\top}{\vec{v}^\top\vec{v}}\right)\vec{y}. \tag{110}$$

Hence, back to our case, the Householder reflector F and thus be selected as

$$F = I - 2\frac{\vec{v}\vec{v}^\top}{\vec{v}^\top\vec{v}}. \tag{111}$$

3/1 Lecture

Note that one could also reflect $\vec{x}$ to $-\|\vec{x}\|\vec{e}_1$ instead of reflecting to $\|\vec{x}\|\vec{e}_1$. However, we should always choose the one that moves $\vec{x}$ a longer distance for the reason of numerical stability. This can be simply done. For instance, if $x_1 \geq 0$ we can choose to reflect to $-\|\vec{x}\|\vec{e}_1$, and otherwise we can choose to reflect to $\|\vec{x}\|\vec{e}_1$. Now that we have

$$Q_n \cdots Q_2 Q_1 A = R, \tag{112}$$

then it suffices to construct Q by $Q = Q_1^\top Q_2^\top \cdots Q_n^\top = Q_1 Q_2 \cdots Q_n$ (recall that Q_k are orthogonal, and by construction symmetric). What would the algorithm be exactly, and what is the runtime?

5.9 The Singular Value Decomposition

Let $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $\operatorname{rank}(A) = n$. Then A can be factorized as

$$A = USV^\top, \tag{113}$$

where $U \in \mathbb{R}^{m\times n}$ is orthogonal (*i.e.*, with orthonormal columns), $S \in \mathbb{R}^{n\times n}$ is diagonal with entries $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$, and $V^\top \in \mathbb{R}^{n\times n}$ is orthogonal as well. We can think of U and $V^\top$ as rotation and reflection, and S as positive scaling. Mathematically, the singular value decomposition can be computed by observing that $A^\top A$ is positive semi-definite and therefore can be written as

$$A^\top A = VS^2V^\top, \tag{114}$$

according to *eigen-decomposition*. Then, by setting $U = AVS^{-1}$, we will see that $US = AV$, and thus $A = USV^\top$. However, this is numerically more unstable than dealing with A directly since

$$K(A^\top A) = K^2(A). \tag{115}$$

Instead, observe (assuming $m = n$) that the matrix

$$H = \begin{pmatrix} 0 & A^\top \\ A & 0 \end{pmatrix} \in \mathbb{R}^{2m\times 2m} \tag{116}$$

is orthogonal, and that if $A = USV^\top$, we will be able to deduce that

$$H \begin{pmatrix} V & V \\ U & -U \end{pmatrix} = \begin{pmatrix} V & V \\ U & -U \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & -S \end{pmatrix}. \tag{117}$$

This does not involve matrix squaring, but deals with a $2m \times 2m$ matrix instead of a $m \times n$ one. We will introduce more on how to efficiently compute singular value decompositions later.

Pseudoinverse

Recall that for $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $\operatorname{rank}(A) = n$, the solution to the least-squares problem must satisfy the normal equations, that is,

$$A^\top A\vec{x} = A^\top \vec{b} \implies \vec{x} = (A^\top A)^{-1}A^\top \vec{b}. \tag{118}$$

The matrix $(A^\top A)^{-1}A^\top$ is known as the *pseudoinverse* of A, which is commonly denoted by A^+. Substituting A with its singular value decomposition, we have that

$$A^+ = (A^\top A)^{-1}A^\top = (VS^2V^\top)^{-1}(USV^\top)^\top = VS^{-1}S^{-1}V^{-1}VSU^\top = VS^{-1}U^\top. \tag{119}$$

In the case that $m > n$, clearly A is not invertible, but with the pseudoinverse, we have that

$$A^+A = VS^{-1}U^\top USV^\top = VS^{-1}SV^\top = VV^\top = I. \tag{120}$$

Midterm Review

- **Bisection:** Linear convergence, absolute error goes down by half in each iteration.
- **Secant method:** Converges with $\alpha \approx 1.62$, using the update formula

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}. \tag{121}$$

- **Newton's method:** Quadratic convergence, using the update formula

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{122}$$

  Note that Newton's method may fail because the first derivative evaluates to zero. It may also fail to converge for certain initial guesses. As for the multivariate version, the update step rule is

$$\vec{x}_{k+1} = \vec{x}_k - J^{-1}(\vec{x}_k)\vec{f}(\vec{x}_k), \tag{123}$$

  where the (i, j)th entry of the Jacobian matrix is $J_{ij} = \partial f_i / \partial x_j$ (each row is an f_i, each column is an x_j).
- **Asymptotic rate of convergence:** Sometimes called *order*, defined as

$$\rho = -\log_{10} \mu. \tag{124}$$

  Note that this is defined only for $\alpha = 1$, *i.e.*, linear convergence.
- **Induced matrix norms:** The induced l_1 norm of A satisfies

$$\|A\|_1 = \max_{j=1,\cdots,n} \sum_{i=1}^{n} |A_{ij}|. \tag{125}$$

  The induced l_2 norm of A satisfies

$$\|A\|_2 = \sqrt{\max_{j=1,\cdots,n} \lambda_j}, \tag{126}$$

  where λ_j are the eigenvalues of $A^\top A$. In other words, the induced l_2 norm of A is its largest singular value. Moreover, note that the induced l_2 norm of A^{-1} is the inverse of the smallest singular value of A.
- **Condition number:** The absolute condition number of A is $\|A^{-1}\|$, and relative $\|A^{-1}\| \cdot \|A\|$. Moreover, note that $K(A^\top A) = K^2(A)$.
- **Convexity:** f is convex on $[a, b]$ if for any $x, y \in [a, b]$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha) f(y). \tag{127}$$

  Moreover, f is convex if its Hessian matrix of second-order derivatives is *positive semidefinite*, *i.e.*, symmetric with all eigenvalues real and positive.
- **Optimization:** The update formula is

$$\vec{x}_{k+1} = \vec{x}_k - H^{-1}(\vec{x}_k)\vec{\nabla} f(\vec{x}_k). \tag{128}$$

  However, explicit Hessian matrices are expensive to compute, so we use the *quasi-Newton* method, such that

$$H_{k+1} = H_k - \frac{H_k \vec{s}_k \vec{s}_k^\top H_k}{\vec{s}_k^\top H_k \vec{s}_k} + \frac{\vec{y}_k \vec{y}_k^\top}{\vec{y}_k^\top \vec{s}_k}, \qquad \vec{s}_k = \vec{x}_{k+1} - \vec{x}_k, \quad \vec{y}_k = \vec{\nabla} f(\vec{x}_{k+1}) - \vec{\nabla} f(\vec{x}_k). \tag{129}$$

- **Sherman-Morrison-Woodbury:** It states that

$$(A + \vec{u}\vec{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\vec{u}\vec{v}^\top A^{-1}}{1 + \vec{v}^\top A^{-1}\vec{u}}. \tag{130}$$

  With this formula, the update of Hessian matrix inverse can be written as

$$H_{k+1}^{-1} = \left(I - \frac{\vec{s}_k \vec{y}_k^\top}{\vec{y}_k^\top \vec{s}_k}\right) H_k^{-1} \left(I - \frac{\vec{y}_k \vec{s}_k^\top}{\vec{y}_k^\top \vec{s}_k}\right) + \frac{\vec{s}_k \vec{s}_k^\top}{\vec{y}_k^\top \vec{s}_k}. \tag{131}$$

- **Real symmetric matrices:** Such matrices can be diagonalized as $A = P^\top DP$, where D is a diagonal matrix of eigenvalues and P is orthogonal, *i.e.*, with orthonormal columns so that $P^\top = P^{-1}$.
- **LU factorization:** $A = LU$, where L is lower triangular and U is upper triangular. This is done similar to Gaussian elimination, by multiplying elementary row operation matrices on the left of A.
- **Cholesky factorization:** If A is symmetric positive definite, then we can write $A = U^\top U$, where U is upper (lower) triangular.
- **Residual:** The residual $\vec{r} = \vec{b} - A\tilde{x}$, where $\tilde{x}$ is the computed solution of $A\vec{x} = \vec{b}$.
- **QR factorization:** $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $\text{rank}(A) = n$. We want $\vec{x}$ that minimizes the l_2 norm of the residual, and this minimizer must satisfy the normal equations $A^\top A\vec{x} = \vec{A}^\top\vec{b}$. We can write $A = QR$, where Q is orthogonal (*i.e.*, $Q^\top Q = I$) and R is upper triangular.
- **Singular value decomposition:** $A = USV^\top$, where $U \in \mathbb{R}^{m\times n}$ is orthogonal, $S \in \mathbb{R}^{n\times n}$ is diagonal with entries as singular values, and $V \in \mathbb{R}^{n\times n}$ also orthogonal.

3/6 Lecture

6 Eigenvalue Problems

Recall that $\lambda, \vec{v}$ is an eigenvalue-eigenvector pair of a matrix A if $A\vec{v} = \lambda\vec{v}$. The direct computation would be via the characteristic equations, that is,

$$p(\lambda) = \det(A - \lambda I) = 0. \tag{132}$$

However, p would be a polynomial of degree n if $A \in \mathbb{R}^{n\times n}$. This is extremely expensive: forming p would cost $n!$ flops, and then a nonlinear root finding algorithm must be used to solve $p(\lambda) = 0$. Before introducing cheaper computation methods, we first see its application in solving systems of linear initial value problems.

6.1 Solving Systems of Linear Initial Value Problems

Consider the initial value problem $\vec{y}' = A\vec{y}$, where we have that

$$\vec{y}(t) = \begin{pmatrix} y_1(t) \\ \vdots \\ y_n(t) \end{pmatrix}, \qquad \vec{y}'(t) = \begin{pmatrix} y_1'(t) \\ \vdots \\ y_n'(t) \end{pmatrix} \tag{133}$$

We will ignore the given initial value $(t_0, \vec{y}(t_0))$ for now. One solution to solve this problem is to diagonalize A, such that $A = PDP^{-1}$ with D being the diagonal matrix of eigenvalues and each column of P being an eigenvector corresponding to the eigenvalue (with the same order). Then, we have that

$$\vec{y}' = PDP^{-1}\vec{y} \implies P^{-1}\vec{y}' = DP^{-1}\vec{y}. \tag{134}$$

By denoting $\vec{u} = P^{-1}\vec{y}$, we have that $\vec{u}' = P^{-1}\vec{y}'$ (since this is taking derivative with respect to t), and thus the above equation can be rewritten as

$$\vec{u}' = D\vec{u} \implies \begin{cases} u_1' = \lambda_1 u_1, \\ \quad\vdots \\ u_n' = \lambda_n u_n. \end{cases} \tag{135}$$

This solves to

$$u_i = c_i e^{\lambda_i t}, \qquad u_i' = \lambda_i c_i e^{\lambda_i t}, \tag{136}$$

where c_i can be easily determined by the given initial values (which we will ignore here). Changing the variables back, we can obtain the solution by $\vec{y} = P\vec{u}$. Therefore, we can see that the problem of solving a system of linear initial value problems can be reduced to finding eigenvalues and eigenvectors of A.

6.2 The Gerschgorin Theorems

Theorem 6.1 (Gerschgorin)**.** Let $n \geq 2$ and $A \in \mathbb{C}^{n\times n}$, then all eigenvalues of the matrix A must lie in the region $D = \bigcup_{i=1}^{n} D_i$, where each D_i is the *Gerschogorin disc* defined as the closed circular region

$$D_i = \{z \in \mathbb{C};\ |z - a_{ii}| \leq R_i\}, \qquad R_i = \sum_{j\neq i} |a_{ij}|. \tag{137}$$

Proof. Suppose that $\lambda \in \mathbb{C}$ and $\vec{v} \in \mathbb{C}^n$ are an eigenvalue and the corresponding nonzero eigenvector of A, so that

$$(A\vec{v})_i = \sum_{j=1}^{n} a_{ij} v_j = \lambda v_i. \tag{138}$$

Suppose that v_k is the component of $\vec{v}$ that is largest (or one of the largest) in modulus. Then we have that

$$|\lambda - a_{kk}||v_k| = |\lambda v_k - a_{kk} v_k| = \left|\sum_{j=1}^{n} a_{kj} v_j - a_{kk} v_k\right| = \left|\sum_{j\neq k} a_{kj} v_j\right| \leq \sum_{j\neq k} |a_{kj}||v_j|. \tag{139}$$

Dividing both sides by $|v_k|$, we thus obtain that

$$|\lambda - a_{kk}| \leq \sum_{j\neq k} |a_{kj}| \frac{|v_j|}{|v_k|} \leq \sum_{j\neq k} |a_{kj}| = R_k. \tag{140}$$

Therefore, $\lambda \in D_k$ assuming v_k is the component of $\vec{v}$ that is largest (or one of the largest) in modulus. Hence, we can guarantee that $\lambda \in \bigcup_{i=1}^{n} D_i = D$, so the proof is complete. □

Theorem 6.2 (Gerschgorin's second theorem)**.** In the above scenario, if m disks are connected, then there are m eigenvalues in that connected region.

We will note prove this theorem now, but revisit it in detail when discussing Jacobi's method later.

6.3 The Power Method

We want to calculate the eigenvalue with the largest magnitude and its associated eigenvector (assuming that the matrix A is diagonalizable). We start with a random vector $\vec{w}$, which must be a linear combination of the eigenvectors of A, such that

$$\vec{w} = \sum_{j=1}^{n} c_j \vec{v}_j. \tag{141}$$

Applying A to $\vec{w}$, we will obtain that

$$A\vec{w} = A\left(\sum_{j=1}^{n} c_j \vec{v}_j\right) = \sum_{j=1}^{n} c_j A\vec{v}_j = \sum_{j=1}^{n} c_j \lambda_j \vec{v}_j. \tag{142}$$

Now, if we apply another A to this resulting vector, we can see that

$$A^2\vec{w} = A\left(\sum_{j=1}^{n} c_j \lambda_j \vec{v}_j\right) = \sum_{j=1}^{n} c_j \lambda_j A\vec{v}_j = \sum_{j=1}^{n} c_j \lambda_j^2 \vec{v}_j \tag{143}$$

Therefore, for sufficiently large k, we will have that

$$A^k\vec{w} = \sum_{j=1}^{n} c_j \lambda_j^k \vec{v}_j, \tag{144}$$

which will be dominated by $c_1\lambda_1^k\vec{v}_1$, assuming that $|\lambda_1| > |\lambda_2| > |\lambda_3| > \cdots$ (*i.e.*, λ_1 is the largest eigenvalue of A in modulus). So eventually, if $\vec{y}^{(k)} = A^k\vec{w}$, then $\vec{y}^{(k+1)} = A^{k+1}\vec{w} = A\vec{y}^{(k)} \approx \lambda_1\vec{y}^{(k)}$. Now, normalizing iterants in every step, we will have that

$$\vec{w}_0 = \frac{\vec{w}}{\|\vec{w}\|}, \qquad \vec{w}_1 = \frac{A\vec{w}_0}{\|A\vec{w}_0\|}, \qquad \cdots, \qquad \vec{w}_k = \frac{A\vec{w}_{k-1}}{\|A\vec{w}_{k-1}\|}. \tag{145}$$

Under this normalization, the largest eigenvalue λ_1 is approximately equal to

$$\lambda_1 = \frac{w_{k,i}}{w_{(k-1),i}}, \tag{146}$$

where $w_{k,i}$ represents the ith component of $\vec{w}_k$. Furthermore, we can in fact obtain a better estimation of λ_1 by

$$\lambda_1 = \langle A\vec{w}_k, \vec{w}_k \rangle, \tag{147}$$

since $A\vec{w}_k \approx \lambda_1\vec{w}_k$, which means that $\vec{w}_k^\top A\vec{w}_k = \lambda_1\vec{w}_k^\top\vec{w}_k = \lambda_1$ given $\vec{w}_k$ is already normalized. Finally, the $\vec{w}_k$'s will approach $\vec{v}_1$ as $k \to \infty$.

Rate of convergence. If k is sufficiently large, then $A^k\vec{w} = c_1\lambda_1^k\vec{v}_1$ (assuming $c_1 > 0$). Then we can compute that

$$\vec{v}_1 \approx \frac{1}{c_1\lambda_1^k}A^k\vec{w} = \frac{1}{c_1\lambda_1^k}\sum_{j=1}^{n} c_j\lambda_j^k\vec{v}_j = \vec{v}_1 + \frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\vec{v_2} + \frac{c_3}{c_1}\left(\frac{\lambda_3}{\lambda_1}\right)^k\vec{v_3} + \cdots + \frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\vec{v_n}. \tag{148}$$

Since $|\lambda_j| < |\lambda_1|$ for $j \geq 2$, we know that $(\lambda_j/\lambda_1)^k \to 0$ as $k \to \infty$. Since the power method normalizes the iterants $\vec{w}_k$ in every step and we assume that the eigenvectors have unit norm, clearly $\vec{w}_k$ converges to $\vec{v}_1$ as $k \to \infty$. Moreover, we can see that

$$\|\vec{w}_k - \vec{v}_1\| \approx \left|\frac{c_2}{c_1}\right|\left|\frac{\lambda_2}{\lambda_1}\right|^k + \texttt{lower order terms} \sim O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^k\right), \tag{149}$$

so that the rate of convergence of the power method depends on the gap in the eigenvalues, specifically the relative size of λ_2 to λ_1. In other words, if $|\lambda_1| \gg |\lambda_2|$, the power method converges fast but if $|\lambda_1| \approx |\lambda_2|$, the convergence would be very slow. We would need certain techniques to guarantee a sufficiently large gap between the eigenvalues.

The power method with shift. If a matrix A has eigenvalues $\lambda_1, \cdots, \lambda_n$, then the matrix $A - sI$ would have eigenvalues $\lambda_1 - s, \cdots, \lambda_n - s$. The reason is, if $\vec{v}$ is an eigenvector of A corresponding to the eigenvalue λ, then

$$(A - sI)\vec{v} = A\vec{v} - s\vec{v} = \lambda\vec{v} - s\vec{v} = (\lambda - s)\vec{v}. \tag{150}$$

Based on this observation, we can use the power method on $A - sI$ instead of A, so that we can choose s to increase the convergence rate. How to choose s? Under which conditions is the shifted power method useful? Power method with shift allows for computing the most negative or the most positive eigenvalue.

3/20 Lecture

The inverse power method with shift. How do we compute eigenvalues in the middle, *i.e.*, if $\lambda_1 > \lambda_2 > \cdots > \lambda_{n-1} > \lambda_n$, then the power method with shift can compute either λ_1 or λ_n. To compute $\lambda_2, \cdots, \lambda_{n-1}$, we would need a different algorithm. The idea is to apply the power method to find eigenvalues of $(A - sI)^{-1}$, which is called the *inverse power method with shift*. If A has some eigenvalue λ, then A^{-1} has the eigenvalue λ^{-1}. Furthermore, $(A - sI)^{-1}$ should have the eigenvalue $(\lambda - s)^{-1}$. If we choose s properly to make $(\lambda - s)^{-1}$ large, then the inverse power method with shift can converge very rapidly.

Choose s close to λ_l so that $(\lambda_l - s)^{-1}$ becomes very large in absolute value, while $(\lambda_j - s)^{-1}$ for $j \neq l$ remains bounded. This scheme is of course much more expensive since applying A^{-1} requires solving a linear system, which takes $\Theta(n^3)$ flops (versus $\Theta(n^2)$ for the power method). The algorithm can be written as follows.

1: set $\vec{w}_0$ to be random;
2: solve $(A - sI)\vec{y}_1 = \vec{w}_0$, which is equivalent to $\vec{y}_1 = (A - sI)\vec{w}_0$;
3: set $\vec{w}_1 = \vec{y}_1 / \|\vec{y}_1\|$;
4: proceed as in the power method;

The hard part is knowing what to choose for s, which would require estimates for the eigenvalues. Moreover, both the power method with shift and the inverse power method with shift compute only one pair of eigenvalue and eigenvector at a time.

6.4 The Jacobi Method

Can we compute all the eigenvalues and their corresponding eigenvectors at the same time? Note that if A were diagonal, then we immediately know its eigenvalues, so the problem is how to make A diagonal. Recall the *similarity transform*, which states that if $B = M^{-1}AM$, then B is similar to A, *i.e.*, they have the same eigenvalues.

Proof. Look at the characteristic polynomial, such that

$$\begin{aligned} \rho_B(\lambda) &= \det(B - \lambda I) = \det(M^{-1}AM - \lambda I) = \det(M^{-1}AM - \lambda M^{-1}M) \\ &= \det(M^{-1}(A - \lambda I)M) = \det(M^{-1})\det(A - \lambda I)\det(M) = \rho_A(\lambda). \end{aligned} \tag{151}$$

Since eigenvalues are just the roots of the characteristic polynomials, the proof is complete. □

Note that if M were chosen to be the matrix of eigenvectors of A, then $A = MDM^{-1}$ with D being the diagonal matrix of eigenvalues. Then, $D = M^{-1}AM$ would be the diagonalization of A.

Example 6.3. Let A be a real symmetric 2×2 matrix, such that

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}. \tag{152}$$

Real symmetric matrices have real eigenvalues, and are in fact *orthogonally diagonalizable*, *i.e.*, can be diagonalized by orthogonal matrices. Note that all 2×2 orthogonal matrices can be parametrized as

$$V = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}, \tag{153}$$

i.e., 2×2 rotation matrices. We thus want

$$V^\top AV = V^{-1}AV = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}. \tag{154}$$

By expanding the expression, we have that

$$\begin{aligned} &\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} a & b \\ b & d \end{pmatrix} \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \\ \implies &\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} a\cos\phi - b\sin\phi & a\sin\phi + b\cos\phi \\ b\cos\phi - d\sin\phi & b\sin\phi + d\cos\phi \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}. \end{aligned} \tag{155}$$

By comparing the diagonal entries, we have that

$$a\sin\phi\cos\phi + b\cos^2\phi - b\sin^2\phi - d\sin\phi\cos\phi = 0, \tag{156}$$

$$a\sin\phi\cos\phi - b\sin^2\phi + b\cos^2\phi - d\sin\phi\cos\phi = 0. \tag{157}$$

The two equations above are exactly the same. We rewrite any one of them to obtain that

$$(a - d)\sin\phi\cos\phi + b(\cos^2\phi - \sin^2\phi) = 0 \implies \frac{a - d}{2}\sin 2\phi + b\cos 2\phi = 0. \tag{158}$$

This implies that, when $d - a \neq 0$, ϕ is given by

$$\tan 2\phi = \frac{2b}{d - a} \implies \phi = \frac{1}{2}\arctan\left(\frac{2b}{d - a}\right). \tag{159}$$

If $d - a = 0$, then clearly $\cos 2\phi = 0$ (as long as A is not diagonal, *i.e.*, $b \neq 0$). This would give $\phi = \pi/4$. In C or Fortran we use `phi = atan(d-a,2b) / 2`. Up till now, we have determined ϕ, such that

$$V^\top A V = D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}. \tag{160}$$

The Jacobi method for an $n \times n$ matrix. Define

$$R^{pq}(\phi) = \begin{array}{c} \left(\begin{array}{c|c|c|c|c} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ \hline & & \cos\phi & & \sin\phi \\ \hline & & & 1 & \\ & & & & \ddots \\ & & & & & 1 \\ \hline & & -\sin\phi & & \cos\phi \\ \hline & & & & & 1 \\ & & & & & & 1 \\ & & & & & & & 1 \end{array}\right) \begin{array}{l} \\ \\ \\ \leftarrow \texttt{row p} \\ \\ \\ \\ \\ \leftarrow \texttt{row q} \\ \\ \\ \\ \end{array} \\ \begin{array}{ccccccc} & & \uparrow & & & \uparrow & \\ & & \texttt{col p} & & & \texttt{col q} & \end{array} \end{array} \tag{161}$$

$R^{pq}(\phi)$ can be used to set the (p,q)th and (q,p)th entries of a real symmetric matrix A to zero. Moreover, the transformation $R^{pq}(\phi)^\top A R^{pq}(\phi)$ leaves all rows and columns unchanged except for the pth row and column and the qth row and column of A. The algorithm is given as follows.

set $A^{(0)} = A$;
$k \leftarrow 0$;
repeat
  find the (p,q)th element in $A^{(k)}$ with maximum absolute value with $p \neq q$;
  compute $\phi_k = \arctan\left(2a_{pq}^{(k)}/(a_{qq}^{(k)} - a_{pp}^{(k)})\right)$;
  set $A^{(k+1)} = R^{pq}(\phi_k)^\top A^{(k)} R^{pq}(\phi_k)$;
  $k \leftarrow k+1$;
until all entries $|a_{ij}^{(k)}| < \epsilon$, $i \neq j$;

The reason this algorithm works is that

$$A^{(k)} \to \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} \qquad \text{as } k \to \infty. \tag{162}$$

To summarize, we apply a sequence of $R(\phi_k)$'s to A that zero out all off-diagonal elements. Furthermore, this sequence will converge to the matrix of eigenvectors.

Convergence of the Jacobi method. The question now is, do elements that are set to zero stay zero forever in the Jacobi method? The answer is no. The idea behind convergence here is, every Jacobi rotation (*i.e.*, applying $R^\top$ and R) moves the "mass" of the matrix from off-diagonal positions to diagonal positions. We first state a lemma.

Lemma 6.4. If R is an orthogonal transformation and $A^\top = A$, then the Frobenius norms $\|A\|_F = \|R^\top A R\|_F$.

Proof. Recall that the Frobenius norm is defined as

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2}. \tag{163}$$

Let $B = R^\top AR$, then A and B have the same eigenvalues, and $B^2 = (R^\top AR)(R^\top AR) = R^\top A^2 R$, implying that A^2 and B^2 also have the same eigenvalues. Therefore, $\operatorname{tr}(A^2) = \operatorname{tr}(B^2)$. However, note that

$$\|A\|_F^2 = \operatorname{tr}(A^\top A) = \operatorname{tr}(A^2) = \operatorname{tr}(B^2) = \|B\|_F^2, \tag{164}$$

so the proof is complete. The fact that $\|A\|_F^2 = \operatorname{tr}(A^\top A)$ is already prove in some previous Homework. □

3/22 Lecture

Now we split the Frobenius norm into a diagonal piece and an off-diagonal piece, such that

$$S(A) =: \|A\|_F^2 = \sum_{i,j} |a_{ij}|^2 = \sum_i |a_{ii}|^2 + \sum_{i \neq j} |a_{ij}|^2 =: D(A) + L(A). \tag{165}$$

Theorem 6.5. Let $A^{(k)}$ be the kth iterant in the Jacobi algorithm, then

$$\lim_{k\to\infty} L(A^{(k)}) = 0, \qquad \lim_{k\to\infty} D(A^{(k)}) = \operatorname{tr}(A^2). \tag{166}$$

Proof. Let a_{pq} be the off-diagonal element of A with the largest absolute value. Let $B = R^{pq}(\phi)^\top AR^{pq}(\phi)$, a single Jacobi rotation. Then we can compute that

$$\underbrace{\begin{pmatrix} b_{pp} & b_{pq} \\ b_{qp} & b_{qq} \end{pmatrix}}_{\tilde{B}} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}^\top \underbrace{\begin{pmatrix} a_{pp} & a_{pq} \\ a_{qp} & a_{qq} \end{pmatrix}}_{\tilde{A}} \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}, \tag{167}$$

and do not forget that $b_{pq} = b_{qp} = 0$ by construction (*i.e.*, taking the suitable ϕ value). But from the lemma, we see that $\|B\|_F^2 = \|A\|_F^2$, so that $b_{pp}^2 + b_{qq}^2 = a_{pp}^2 + 2a_{pq}^2 + a_{qq}^2$. Now since $S(A) = S(B)$ as in the previous lemma, we have that $D(A) + L(A) = D(B) + L(B)$. Also note that the diagonal entries of B are the same as those of A, except the ones in rows p and q. Therefore, we can deduce that

$$D(B) - D(A) = (b_{pp}^2 + b_{qq}^2) - (a_{pp}^2 + a_{qq}^2) = 2a_{pq}^2 \implies D(B) = D(A) + 2a_{pq}^2. \tag{168}$$

Consequently, we also have that

$$L(B) = L(A) - 2a_{pq}^2. \tag{169}$$

Now since a_{pq} was the largest off-diagonal element of A, we have that

$$L(A) \leq n(n-1)a_{pq}^2 \implies a_{pq}^2 \geq \frac{L(A)}{n(n-1)}. \tag{170}$$

Therefore, we can see that

$$L(B) = L(A) - 2a_{pq}^2 \geq L(A) - \frac{2L(A)}{n(n-1)} = L(A)\left(1 - \frac{2}{n(n-1)}\right). \tag{171}$$

Now we relabel the matrices such that $A^{(0)} = A$ and $A^{(1)} = B$, then

$$L(A^{(1)}) \leq L(A^{(0)})\left(1 - \frac{2}{n(n-1)}\right), \qquad L(A^{(2)}) \leq L(A^{(0)})\left(1 - \frac{2}{n(n-1)}\right)^2, \qquad \text{etc.} \tag{172}$$

After k iterations, we will thus obtain that

$$L(A^{(k)}) \leq L(A^{(0)})\left(1 - \frac{2}{n(n-1)}\right)^k. \tag{173}$$

This implies that $L(A^{(k)}) \to 0$ as $k \to \infty$. Moreover, since $S(A^{(k)}) = \operatorname{tr}(A^2)$, we can easily see that $D(A^{(k)}) \to \operatorname{tr}(A^2)$ as $k \to \infty$. This completes the proof. □

Applying the Gerschgorin theorem. $A^{(k)}$ and A have the same eigenvalues, and $L(A^{(k)}) \to 0$ as $k \to \infty$. Therefore, the Gerschgorin disks of $A^{(k)}$ have radii going towards 0 as well. Therefore, the eigenvalues of $A^{(k)}$, and thus of A, are the limit of the diagonal of $A^{(k)}$.

Rate of convergence. We have shown that

$$L(A^{(k)}) \leq L(A^{(0)}) \left(1 - \frac{2}{n(n-1)}\right)^k. \tag{174}$$

If $n = 1000$, then $1 - \frac{2}{n(n-1)} \approx 0.99999799799\cdots$. In this case, even if $k = 10000$, $\left(1 - \frac{2}{n(n-1)}\right)^k$ is just approximately 0.98, far from 0. However, real-life convergence is often *much* faster than indicated in the proof. One final remark: the Jacobi method can be terminated when $L(A^{(k)}) < \epsilon$. Then we have that

$$A^{(k)} = \underbrace{R(\phi_k)^\top \cdots R(\phi_1)^\top}_{R^\top} A \underbrace{R(\phi_1) \cdots R(\phi_k)}_{R} \approx \texttt{diag}. \tag{175}$$

Then, $A = RA^{(k)}R^\top$. We say that R diagonalizes A. Here R is the matrix in which each column is the approximate of an eigenvector of A, and $A^{(k)}$ is approximately a diagonal matrix, whose diagonal entries are approximately the eigenvalues of A (in the same order as the eigenvectors).

6.5 The QR Method

For general matrices, the *QR method* can be used to find all eigenvalues. We will examine the algorithm first, and then analyze it. The QR algorithm is described as follows.

set $A^{(0)} = A$;
$k \leftarrow 1$;
repeat
  factorize $A^{(k-1)} = Q^{(k)}R^{(k)}$;
  compute $A^{(k)} = R^{(k)}Q^{(k)}$;
  $k \leftarrow k + 1$;
until the stopping criterion is met;

Under certain assumptions, $A^{(k)}$ would converge to some upper triangular matrix. Furthermore, we have that

$$A^{(k)} = R^{(k)}Q^{(k)} = (Q^{(k)})^{-1}A^{(k-1)}Q^{(k)}, \tag{176}$$

so $A^{(k)}$ and $A^{(k-1)}$ have the same eigenvalues, implying that $A^{(k)}$ and A also have the same eigenvalues via iterations. Three things must be done in order for this to be a usable and accelerated algorithm.

(1) First, reduce A to tridiagonal (*i.e.*, the only nonzero entries must be on the main diagonal, the subdiagonal right below that, and the subdiagonal right above that). This can also be called *Hessenburg* (*i.e.*, both *upper Hessenburg* and *lower Hessenburg*). We do this by using the Householder reflections.

(2) Apply to the shifted matrices $A^{(k)} - \mu^{(k)}I$, with $\mu^{(k)}$ an estimate of some λ.

(3) Use a deflation, *i.e.*, decouple into smaller subproblems.

To understand the "pure" QR algorithm above, we first study the subspace iteration (essentially the power method on groups of vectors), which is described as follows.

To be done...

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